

Fabio Arias

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PROFESSIONAL SUMMARY

Full Stack Developer with over 5 years of experience in software development, specialized in scalable architectures, microservices, and cloud technologies. Expert in the JavaScript/TypeScript ecosystem (Node.js, React, Angular) and PHP (Laravel), with solid experience in Python for AI/ML and microservices.

Experienced technical tutor and instructor, having led more than 20 student cohorts in intensive backend development programs, combining strong technical expertise with exceptional pedagogical skills. Proven track record in hackathons, developing innovative AI-driven solutions (RAG, facial recognition, LLMs) and Azure-based services.

University Degree in Programming completed, with a Final Capstone Project focused on a Training Management System. Demonstrated ability to design and build complex applications from scratch within 24–48 hours under high-pressure, agile, and multidisciplinary team environments.

PROFESSIONAL EXPERIENCE

Senior Consultant | 4Latam

Septiembre 2021 – Present, Buenos Aires, Capital Federal

- Collaborated on public tenders and individual projects for freelance developers, enabling effective connections between clients and technical talent
- Provided technical and strategic consulting in software development, ensuring quality, efficiency, and alignment with client standards
- Supervised deadlines and enforced best coding practices across multidisciplinary teams
- Conducted code reviews and technical audits to ensure maintainable, scalable, and high-quality codebases

Backend Development Instructor & Teaching Assistant | CoderHouse

CoderHouse

Marzo 2021 – Present, Buenos Aires, Argentina

- Led over 20 student cohorts in 6-month intensive programs, delivering comprehensive training in modern web technologies (HTML, CSS, JavaScript, React, Node.js, MongoDB)
- Designed and implemented scalable, secure, and production-ready backend applications with professional authentication and authorization mechanisms
- Designed and optimized MongoDB and MySQL databases, improving query and aggregation performance by up to 40%
- Managed data persistence across multiple databases, ensuring integrity and consistency through transactions and validations
- Supported students end-to-end throughout the programs, achieving an 85% graduation rate
- Mentored students through simultaneous individual projects, overcoming weekly technical challenges and ensuring continuous progress
- Fostered an interactive and collaborative learning environment, adapting teaching methodologies to individual learning needs
- Integrated real-world industry examples and best practices to prepare students for professional careers in web development

Freelance Software Developer

Freelance

Enero 2020 – Present, Formosa, Argentina

- Developed custom web solutions for local companies
- Built management systems, e-commerce platforms, and tailored web applications
- Provided ongoing technical support and maintenance for production systems

PROJECTS

AACOP – Training Management System | University Capstone Project (UTN)

Laravel 12, Livewire 3, TailwindCSS, MySQL | 2024–2025

- Designed and developed a complete web-based system as the Final Capstone Project for the University Degree in Programming
- Built a centralized platform for administrative and academic management of internal training programs, including enrollment, attendance, and certification workflows
- Designed a relational database architecture with five core tables (users, trainings, enrollments, attendance, final_grades), ensuring referential integrity
- Implemented role-based access control (Administrator, Instructor, Participant) with custom permissions and views using Laravel Breeze
- Developed full CRUD functionality for training programs with automatic instructor assignment and capacity validation
- Implemented real-time enrollment validation to prevent duplicates using database-level UNIQUE constraints
- Built an attendance management module allowing instructors to record session-by-session attendance through an intuitive interface
- Implemented a final grading system with approval states and automatic generation of downloadable certificates
- Developed an administrative dashboard with real-time metrics (active enrollments, approval rates, ongoing trainings)
- Leveraged Livewire 3 for reactive components without additional JavaScript, improving UX with real-time updates
- Designed a fully responsive UI using TailwindCSS and Blade Templates
- Implemented migrations and seeders to simplify deployment and testing

- Applied MVC architecture principles and Laravel best practices (Service Providers, Form Requests, Policies)

Project: NeuroDesk | Microsoft Innovation Studio Challenge – Azure AI Hackathon

Backend & Cloud Architecture

- Participated in a technology hackathon developing *NeuroDesk*, an organizational immune system powered by Azure AI for enterprise automation
- Collaborated on backend architecture design and intelligent agent orchestration using Azure OpenAI (GPT-4) and Semantic Kernel
- Implemented automation workflows for IT, Human Resources (RAG), sentiment analysis, NLP processing, and human escalation mechanisms
- Integrated cloud services including Azure Web Apps for Containers, Azure Cosmos DB (Serverless), Azure AI Search, Logic Apps, Automation, Blob Storage, and Content Safety
- Applied scalable, event-driven, and cloud-native architecture principles
- Worked efficiently under tight deadlines, delivering a functional solution within 48 hours

Airport Management Microservices System – Flight

• NestJS, TypeScript, RabbitMQ, Docker, PostgreSQL, Vite.js

- Designed and developed a distributed microservices architecture for an airport management system using asynchronous communication
- Implemented three independent microservices: API Gateway, Flights Service, and Passengers Service, each with its own PostgreSQL database
- Designed a centralized API Gateway for intelligent request routing and service orchestration using the Proxy pattern
- Implemented RabbitMQ-based asynchronous messaging for full service decoupling
- Containerized the entire infrastructure with Docker and Docker Compose for consistent and scalable deployments
- Developed a modern frontend using Vite.js and TypeScript with real-time flight and passenger management
- Implementé base de datos PostgreSQL independiente por microservicio, garantizando aislamiento de datos y escalabilidad horizontal
- Applied Hexagonal Architecture and Domain-Driven Design (DDD) principles
- Configuré health checks, logging centralizado y manejo de errores distribuido para monitoreo de servicios
- Implemented health checks, centralized logging, distributed error handling, retry mechanisms, dead-letter queues, and circuit breakers

Nexo Emprendedores – Digital Ecosystem for Entrepreneurs & Investors

• Instituto Politécnico Científico | 24-hour Hackathon

- Designed and developed a platform connecting entrepreneurs and investors in Argentina’s NEA region, addressing the gap between innovation and funding
- Implemented a Laravel 12 (TALL Stack) main application with PostgreSQL/Supabase
- Built a biometric authentication microservice using Python, FastAPI, face-recognition, and dlib, containerized with Docker
- Developed a full project management system (CRUD) enabling entrepreneurs to showcase initiatives with funding goals and multimedia
- Integrated an AI-powered virtual assistant using RAG with Groq (Llama 3.1), Hugging Face embeddings, and Azure AI Speech for voice interaction
- Implemented dual-role systems (Entrepreneur / Investor) with personalized dashboards and project galleries
- Built real-time widgets for currency exchange rates and regional economic news
- Integrated accessibility features (text resizing, high contrast) following inclusive design standards
- Delivered a fully functional full-stack solution within 24 hours under high-pressure conditions

EDUCATION

University Degree in Programming

National Technological University (UTN) – Resistencia Regional Faculty • Resistencia, Argentina • 2022 – 2025 | Completed

- Final Capstone Project: **AACOP Training Management System**

TECHNICAL SKILLS

Languages:

JavaScript, TypeScript, PHP, Python, C

Frontend:

HTML, CSS, React, Angular, Tailwind CSS, Alpine.js, Livewire

Backend & Frameworks:

Node.js, NestJS, Express, Laravel, FastAPI

Databases:

MongoDB, MySQL, PostgreSQL, Firebase, Redis, Supabase:

DevOps & Cloud:

Docker, Docker Compose, Kubernetes, CI/CD, Git, GitHub

Azure OpenAI, Azure Cosmos DB, Azure AI Search, Azure Web Apps, Azure Logic Apps, Blob Storage, Azure AI Speech

AI & ML:

RAG (Retrieval-Augmented Generation), Groq (Llama 3.1), Hugging Face, Semantic Kernel, face-recognition, dlib, Whisper (STT/TTS)

Other:

GraphQL, SQL, Shell Scripting, PowerShell, Composer, NPM, Yarn, pnpm
Agile / Scrum, Microservices Architecture, Cloud-Native Architecture, Event-Driven Systems
